

CNC Master Reference Handbook

A comprehensive guide for students, operators, and engineers.

1. Essential G-Codes (Categorized)

Motion Commands ●

Code	Function	Explanation	Mill	Lathe
G00	Rapid Positioning	Moves tool rapidly without cutting.	✓	✓
G01	Linear Interpolation	Moves in a straight line at programmed feed rate.	✓	✓
G02	CW Circular Interpolation	Clockwise arc movement.	✓	✓
G03	CCW Circular Interpolation	Counterclockwise arc movement.	✓	✓

Plane Selection & Units ●

Code	Function	Explanation	Mill	Lathe
G17	XY Plane	Selects XY plane for arcs/cycles.	✓	
G18	XZ Plane	Selects XZ plane.	✓	✓
G19	YZ Plane	Selects YZ plane.	✓	
G20	Imperial Units	Sets programming to inches.	✓	✓
G21	Metric Units	Sets programming to millimeters.	✓	✓

Positioning & Offsets ●

Code	Function	Explanation	Mill	Lathe
G40	Cancel Compensation	Turns off cutter radius comp.	✓	✓
G41 / G42	Cutter Comp Left/Right	Offsets path by tool radius.	✓	✓
G43	Tool Length Comp (+)	Applies tool length offset (H).	✓	
G49	Cancel Length Comp	Turns off tool length offset.	✓	
G54-G59	Work Offsets	Selects saved work coordinate origins.	✓	✓
G90	Absolute Positioning	Coordinates are relative to part zero.	✓	✓
G91	Incremental Positioning	Coordinates relative to current position.	✓	✓

Canned Cycles ●

Code	Function	Explanation	Mill	Lathe
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G80	Cancel Canned Cycle	<i>Turns off modal drill/bore cycles.</i>	✓	✓
G81	Drilling Cycle	<i>Basic drill and retract.</i>	✓	✓
G83	Peck Drilling Cycle	<i>Drills and retracts to clear chips.</i>	✓	✓
G84	Tapping Cycle	<i>Feeds in, reverses spindle, feeds out.</i>	✓	✓

2. Essential M-Codes (Machine Control) ●

Code	Function	Explanation
M00	Program Stop	Halts execution until operator presses start.
M01	Optional Stop	Stops if 'Op Stop' button is active on panel.
M03	Spindle Forward	Starts spindle clockwise.
M04	Spindle Reverse	Starts spindle counterclockwise.
M05	Spindle Stop	Halts spindle rotation.
M06	Tool Change	Executes automatic tool change (Mill).
M08 / M09	Coolant ON / OFF	Controls primary flood coolant.
M30	End of Program	Stops machine and rewinds program to top.
M98 / M99	Subprogram Call/Return	Calls another program, then returns to main.

3. Important CNC Concepts

Modal vs. Non-Modal Codes

Modal Codes: Remain active until cancelled or replaced by another code from the same group.

Examples: G90 (Absolute), G17 (XY Plane), G54 (Work Offset).

Non-Modal Codes: Execute only on the line (block) they are programmed on.

Examples: G04 (Dwell), G28 (Return to Home), G53 (Machine Coordinates).

Tool Offsets

H = Tool Length Offset: Associates a length value with the tool.

Example: G43 H01

D = Cutter Radius Comp: Associates a diameter/radius value for profiling.

Example: G41 D01

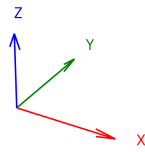
4. Sample CNC Program (Milling)

This is a standard initialization and basic contouring program.

```
%  
O1001                                (Part 1 - Contour)  
G21 G17 G90 G40 G49 G80           (Safety block: Metric, XY, Absolute, Cancel Comps/  
Cycles)  
G54                                  (Select Work Offset 1)  
T1 M06                             (Call Tool 1, Execute Tool Change)  
S1500 M03                          (Spindle Speed 1500 RPM, Forward)  
G00 X0 Y0                          (Rapid move to X0 Y0)  
G43 H01 Z50.0                    (Apply Tool Length Comp 1, Rapid to Z50 clearance)  
G01 Z-5.0 F200                     (Feed into part to Z-5 at 200 mm/min)  
G01 X50.0                          (Cut to X50)  
Y50.0                              (Cut to Y50)  
X0.0                               (Cut to X0)  
Y0.0                               (Cut to Y0)  
G00 Z100.0                         (Rapid retract to Z100)  
M05                                (Spindle Stop)  
M30                                (End of Program / Rewind)  
%
```

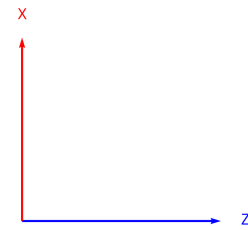
5. Coordinate Systems

Milling (3-Axis)



Milling (3-Axis)

Turning (2-Axis)



Turning (2-Axis)

6. Speeds and Feeds Formulas

Spindle Speed (RPM)

$$\text{RPM} = (1000 \times V_c) / (\pi \times D)$$

Where V_c = Cutting Speed (m/min) and D = Tool Diameter (mm). Use 3.82 instead of 1000 for Imperial (SFM/Inches).

Feed Rate (mm/min or in/min)

$$\text{Feed} = \text{RPM} \times Z \times f_z$$

Where Z = Number of Flutes (Teeth) and f_z = Feed per tooth (chip load).

7. Safety Checklist

- **Always home the machine** before machining.
- **Verify tool offsets** (length and wear) are set correctly.
- **Simulate the program** on the control screen before running.
- **Check work offsets** (G54–G59) are accurately probed.
- **Ensure coolant settings** are aimed correctly at the cutting zone.
- **Confirm spindle direction** (M03 for right-hand tools).
- Keep a hand on the **Feed Hold** button during first article prove-out.